Campus Network Design for Dindori College of Nursing

Objective

The objective of this project is to design and simulate a campus network that connects multiple departments such as Admin, Library, and Lab using routers, switches, and wireless networks. The network uses subnetting with VLSM, Access Control Lists (ACL), wireless security, and monitoring features.

Network Components

- Routers (4)
- Core Switch (1)
- Access Switches (3)
- Wireless Router (1)
- PCs
- Server

Network Topology

The campus network follows a hierarchical design with a core switch connecting all departmental routers. Each router connects to a departmental switch which provides connectivity to PCs and servers.

Departments:

- Admin Department
- Library Department
- Lab Department

IP Addressing Plan (VLSM)

Admin Network: 192.168.10.0/26

Gateway: 192.168.10.1

Library Network: 192.168.10.64/26

Gateway: 192.168.10.65

Lab Network: 192.168.10.128/26

Gateway: 192.168.10.129

Subnet Mask: 255.255.255.192

Cabling

Router to Core Switch – Copper Straight Through

Router to Switch – Copper Straight Through

PC to Switch – Copper Straight Through

Switch to Switch – Copper Crossover

Router Configuration

Admin Router

```
enable
configure terminal
interface g0/0
ip address 192.168.10.1 255.255.255.192
no shutdown

interface g0/1
ip address 10.0.0.1 255.255.255.0
no shutdown
exit
```

Library Router

```
enable
configure terminal
interface g0/0
ip address 192.168.10.65 255.255.255.192
no shutdown

interface g0/1
ip address 10.0.0.2 255.255.255.0
no shutdown
exit
```

Lab Router

```
enable
configure terminal
interface g0/0
ip address 192.168.10.129 255.255.255.192
no shutdown

interface g0/1
ip address 10.0.0.3 255.255.255.0
no shutdown
exit
```

Static Routing Configuration

Admin Router

```
ip route 192.168.10.64 255.255.255.192 10.0.0.2
ip route 192.168.10.128 255.255.255.192 10.0.0.3
```

Library Router

```
ip route 192.168.10.0 255.255.255.192 10.0.0.1
ip route 192.168.10.128 255.255.255.192 10.0.0.3
```

Lab Router

```
ip route 192.168.10.0 255.255.255.192 10.0.0.1
ip route 192.168.10.64 255.255.255.192 10.0.0.2
```

Access Control List (ACL)

ACL is used to restrict Library network from accessing Admin network.

```
access-list 100 deny ip 192.168.10.64 0.0.0.63 192.168.10.0 0.0.0.63
access-list 100 permit ip any any
```

Apply ACL on Library Router:
interface g0/0
ip access-group 100 in

Wireless Configuration
SSID: COLLEGE_WIFI
Security: WPA2-PSK
Password: nursing123

Guest Network
SSID: GUEST_WIFI
Password: guest123

Spanning Tree Protocol
STP prevents switching loops in the network.

enable
configure terminal
spanning-tree mode pvst

SNMP Configuration
Used for network monitoring.
snmp-server community public RO
snmp-server community private RW

PC Configuration Example
Admin PC
IP Address: 192.168.10.2
Subnet Mask: 255.255.255.192
Gateway: 192.168.10.1

Library PC
IP Address: 192.168.10.66
Subnet Mask: 255.255.255.192

Gateway: 192.168.10.65

Lab PC

IP Address: 192.168.10.130

Subnet Mask: 255.255.255.192

Gateway: 192.168.10.129

Testing

Connectivity testing is performed using ping commands.

ping 192.168.10.66

ping 192.168.10.130

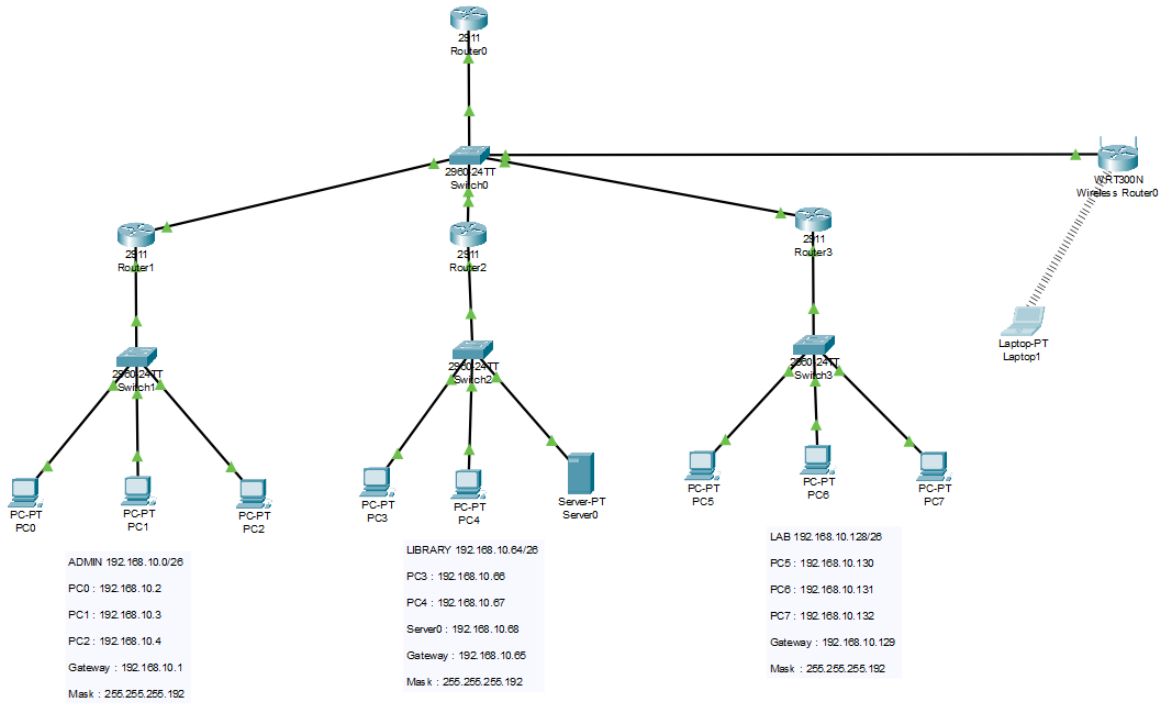
Result

The campus network was successfully simulated. Communication between departments works according to security policies. Library network is restricted from accessing the admin network using ACL while other departments communicate normally.

Conclusion

The project demonstrates how campus networks can be designed using routers, switches, wireless networks, and security mechanisms such as ACL and WPA2 encryption. The use of subnetting improves efficient IP allocation while STP and SNMP improve reliability and monitoring.

Screenshots:



Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#snmp-server community public RO
%SNMP-5-WARMSTART: SNMP agent on host Router is undergoing a warm start
Router(config)#snmp-server community private RW
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown
Router(config-if)#ip route 192.168.10.0 255.255.255.192 10.0.0.1
Router(config)#ip route 192.168.10.64 255.255.255.192 10.0.0.2
Router(config)#


Router con0 is now available


Press RETURN to get started.
```

Copy Paste

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no access-list 100
Router(config)#
Router(config)#access-list 100 deny ip 192.168.10.64 0.0.0.63 192.168.10.0 0.0.0.63
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/1
Router(config-if)#ip access-group 100 out
Router(config-if)#interface g0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#ex
Router(config)#show ip interface brief
                ^
% Invalid input detected at '^' marker.

Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.65   YES manual up                    up
GigabitEthernet0/1     10.0.0.2        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#conf t
%Invalid hex value
Router(config)#ip route 192.168.10.0 255.255.255.192 10.0.0.1
Router(config)#ip route 192.168.10.0 255.255.255.192 10.0.0.1
Router(config)#conf t
%Invalid hex value
Router(config)#no access-list 100
Router(config)#
Router(config)#access-list 100 deny ip 192.168.10.64 0.0.0.63 192.168.10.0 0.0.0.63
Router(config)#access-list 100 permit ip any any
Router(config)#
```

Copy Paste

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#snmp-server community public RO
%SNMP-5-WARMSTART: SNMP agent on host Router is undergoing a warm start
Router(config)#snmp-server community private RW
Router(config)#enable
% Incomplete command.
Router(config)#show ip interface brief
                ^
% Invalid input detected at '^' marker.

Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#show ip interface brief
                ^
% Invalid input detected at '^' marker.

Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown
Router(config-if)#ip route 192.168.10.64 255.255.255.192 10.0.0.2
Router(config)#ip route 192.168.10.128 255.255.255.192 10.0.0.3
Router(config)#show ip route
```

Copy Paste

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#ip route 192.168.10.0 255.255.255.192 10.0.0.1
Router(config)#ip route 192.168.10.64 255.255.255.192 10.0.0.2
Router(config)#ip route 192.168.10.128 255.255.255.192 10.0.0.3
Router(config)#
Router(config)#

|





Router con0 is now available





Press RETURN to get started.
```

Copy Paste

Wireless Router0

Physical | Config | GUI | Attributes

Wireless-N Broadband Router

Firmware Version: v0.93.3

Wireless-N Broadband Router WRT300N

Setup | Setup | Wireless | Security | Access Restrictions | Applications & Gaming | Administration | Status

Basic Setup | DDNS | MAC Address Clone | Advanced Routing

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers)

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP

IP Address: 192 . 168 . 10 . 193

Subnet Mask: 255.255.255.192

DHCP Server Settings

DHCP Server: Enabled Disabled

DHCP Reservation

Start IP Address: 192.168.10. 193

Maximum number of Users: 50

IP Address Range: 192.168.10. 193 - 242

Client Lease Time: 0 minutes (0 means one day)

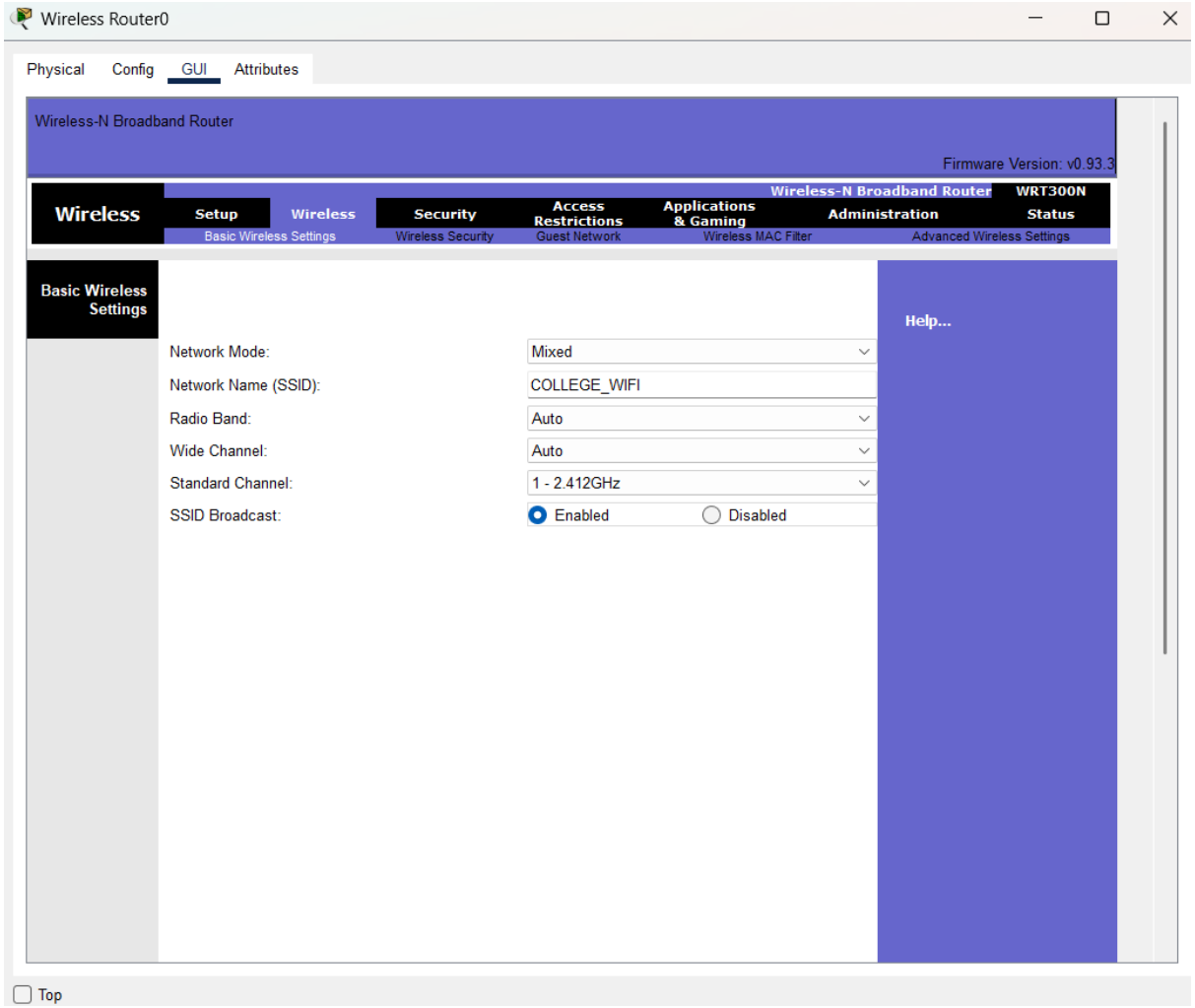
Static DNS 1: 0 . 0 . 0 . 0

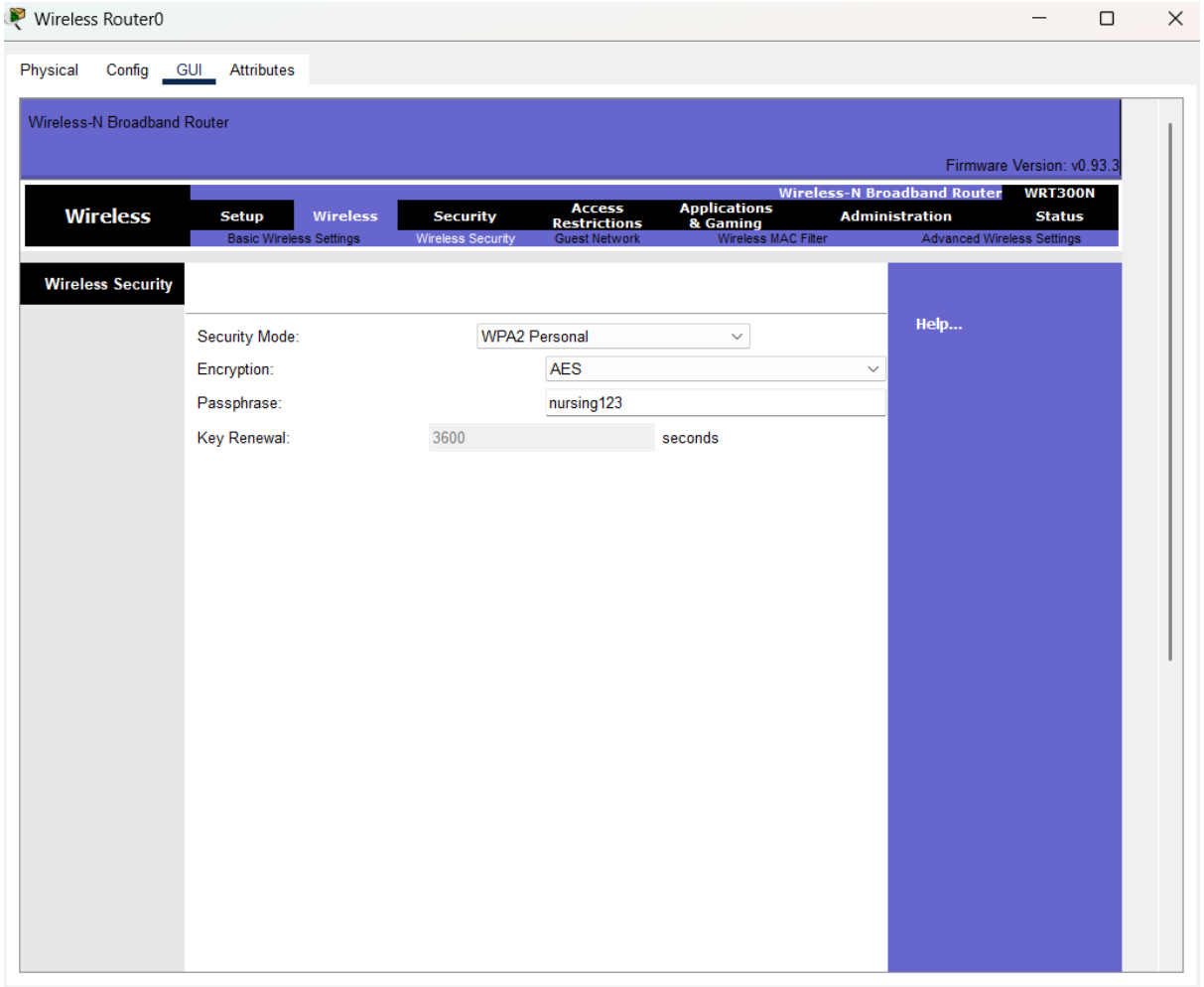
Static DNS 2: 0 . 0 . 0 . 0

Static DNS 3: 0 . 0 . 0 . 0

WINS: 0 . 0 . 0 . 0

Help...


Wireless Router0
Physical
Config
GUI
Attributes
Wireless-N Broadband Router
Firmware Version: v0.93.3
Wireless-N Broadband Router
WRT300N
Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status
Basic Wireless Settings
Wireless Security
Guest Network
Wireless MAC Filter
Advanced Wireless Settings
Basic Wireless Settings
Help...
Network Mode:
Mixed
Network Name (SSID):
COLLEGE_WIFI
Radio Band:
Auto
Wide Channel:
Auto
Standard Channel:
1 - 2.412GHz
SSID Broadcast:
Enabled
Disabled
Top


Wireless Router0
Physical
Config
GUI
Attributes
Wireless-N Broadband Router
Firmware Version: v0.93.3
Wireless-N Broadband Router
WRT300N
Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status
Basic Wireless Settings
Wireless Security
Guest Network
Wireless MAC Filter
Advanced Wireless Settings
Wireless Security
Security Mode:
WPA2 Personal
Encryption:
AES
Passphrase:
nursing123
Key Renewal:
3600
seconds
Help...

PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.65: Destination host unreachable.
Reply from 192.168.10.65: Destination host unreachable.
Reply from 192.168.10.65: Destination host unreachable.
Reply from 192.168.10.65: Destination host unreachable.

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.10.130: bytes=32 time<1ms TTL=126
Reply from 192.168.10.130: bytes=32 time<1ms TTL=126
Reply from 192.168.10.130: bytes=32 time<1ms TTL=126
Reply from 192.168.10.130: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.68

Pinging 192.168.10.68 with 32 bytes of data:

Reply from 192.168.10.68: bytes=32 time<1ms TTL=128
Reply from 192.168.10.68: bytes=32 time<1ms TTL=128
Reply from 192.168.10.68: bytes=32 time<1ms TTL=128
Reply from 192.168.10.68: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Top

PC6

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.129

Pinging 192.168.10.129 with 32 bytes of data:

Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.66

Pinging 192.168.10.66 with 32 bytes of data:

Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.
Reply from 192.168.10.129: Destination host unreachable.

Ping statistics for 192.168.10.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=1ms TTL=126
Reply from 192.168.10.3: bytes=32 time<1ms TTL=126
Reply from 192.168.10.3: bytes=32 time<1ms TTL=126
Reply from 192.168.10.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Top

Laptop1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.66

Pinging 192.168.10.66 with 32 bytes of data:

Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.

Ping statistics for 192.168.10.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.132

Pinging 192.168.10.132 with 32 bytes of data:

Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.
Reply from 192.168.10.193: Destination host unreachable.

Ping statistics for 192.168.10.132:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```